

Docker-01

1. Setup a virtual machine using Ubuntu image.

Create an Ubuntu EC2 instance

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Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type
ami-07a00cf47dbbc844c (64-bit (x86)) / ami-0af878be293432b08 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 26.04, amd64 resolute image

Architecture

AMI ID

Publish Date

Username

64-bit (x86)

ami-07a00cf47dbbc844c

2026-04-21

ubuntu

Verified provider

2. Install docker on ec2.

Update packages: apt update

```
ubuntu@ip-172-31-10-98:~$ sudo apt update
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 Packages [1483 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
```

Install Docker: sudo apt install docker.io -y

```
ubuntu@ip-172-31-10-98:~$ sudo apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils  containerd  dns-root-data  dnsmasq-base  pigz  runc  ubuntu-fan

Suggested packages:
  ifupdown  aufs-tools  cgroupfs-mount  |  cgroup-lite  debootstrap  docker-buildx  docker-compose-v2

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 39
  Download size: 74.0 MB
  Space needed: 281 MB / 48.4 GB available
```

Enable and start Docker:

```
ubuntu@ip-172-31-10-98:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-05-30 15:20:10 UTC; 3min 5s ago
     Invocation: 0b63d9e2468643d385fb64a55388fdb
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2334 (dockerd)
       Tasks: 9
      Memory: 25.2M (peak: 25.5M)
         CPU: 220ms
      CGroup: /system.slice/docker.service
              └─2334 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.746247855Z" level=info msg="Restoring containers: start"
May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.769193048Z" level=info msg="Deleting nftables IPv4 rule"
May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.773687537Z" level=info msg="Deleting nftables IPv6 rule"
May 30 15:20:10 ip-172-31-10-98 dockerd[2334]: time="2026-05-30T15:20:10.961873605Z" level=info msg="Loading containers: done."
```

Check Docker version:

```
ubuntu@ip-172-31-10-98:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-10-98:~$
```

```

ubuntu@ip-172-31-10-98:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-10-98:~$ newgrp docker
ubuntu@ip-172-31-10-98:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:0e760dfdbc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ccl
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:

```

3. Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)

Pull images one by one:

Nginx :docker pull nginx

```

ubuntu@ip-172-31-10-98:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
7f8b1a2b17d8: Pull complete
45381ecb0e2f: Pull complete
b4a248c845e5: Pull complete
5b4d6ff92fc4: Pull complete
5431d0092ffd: Pull complete
830625e1ac85: Pull complete
13fd728be9eb: Pull complete
5e6b66b5e5f1: Download complete
cc5f57206478: Download complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-10-98:~$ 

```

Apache Tomcat : docker pull tomcat

```
ubuntu@ip-172-31-10-98:~$ docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4f4fb700ef54: Pull complete
b40150c1c271: Pull complete
d92a694d8aa0: Pull complete
97c47a5c1a2f: Pull complete
b38893372039: Pull complete
6b82a07ff6e7: Pull complete
65313903a8a8: Pull complete
f80ef884f8c2: Download complete
13883d0c1f97: Download complete
Digest: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
ubuntu@ip-172-31-10-98:~$
```

Ubuntu: docker pull ubuntu

```
ubuntu@ip-172-31-10-98:~$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
ubuntu@ip-172-31-10-98:~$
```

Jenkins: docker pull jenkins

```

ubuntu@ip-172-31-10-98:~$ docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
cebc1e77ceba: Pull complete
3bdbd616c406: Pull complete
6b35bb76b8c8: Pull complete
801814f62c99: Pull complete
c2314d42cca1: Pull complete
76a355a69c2e: Pull complete
5834ae4b8774: Pull complete
ae218d1cfa6e: Pull complete
9014a3401c36: Pull complete
4c140080734e: Pull complete
dd00b4ea1f64: Pull complete
307f8152a55e: Pull complete
653a4e57bccf: Download complete
Digest: sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
ubuntu@ip-172-31-10-98:~$

```

Sonarqube : docker pull sonarqube

```

ubuntu@ip-172-31-10-98:~$ docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
c10b8232386e: Pull complete
f4306f81d85a: Pull complete
907ea4a8bc72: Download complete
8506693bc963: Download complete
Digest: sha256:35bedac3f40cda75969890da59b17d577770844fe6ef659206c678a8e00921c7
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
ubuntu@ip-172-31-10-98:~$

```

Verify images: docker images

```

ubuntu@ip-172-31-10-98:~$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	0e760fd9bc48	25.9kB	9.49kB	U
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	
nginx:latest	5aca99593157	241MB	66MB	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```

ubuntu@ip-172-31-10-98:~$

```

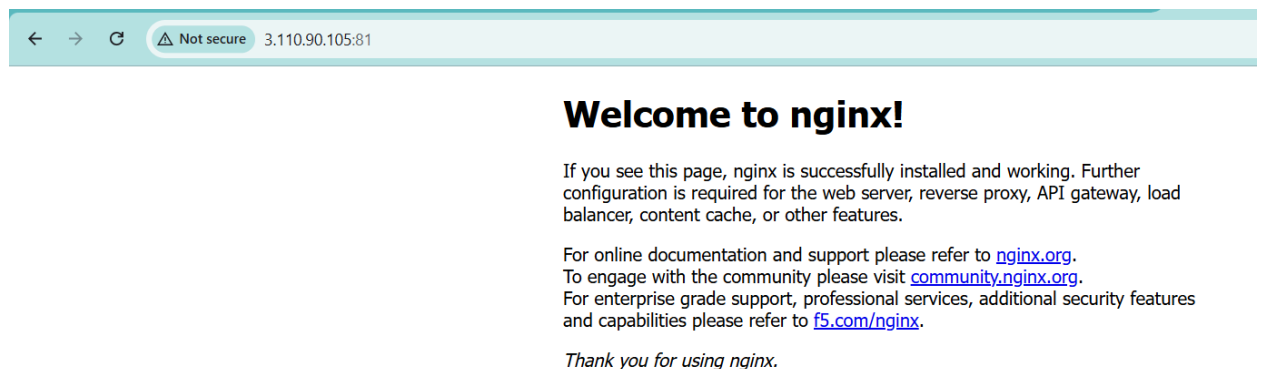
4. Run nginx container and expose on port 81.

Run container: docker run -d --name nginx-container -p 81:80 nginx

Check running containers: docker ps

```
ubuntu@ip-172-31-10-98:~$ docker run -d --name nginx-container -p 81:80 nginx
9dddb9bd4569c5ab8fa7995b2c6e2ef0df1bde03f3f70664c2fbca539bca2642
ubuntu@ip-172-31-10-98:~$
```

Access in browser:



5. Delete the Apache tomcat image from local.

Delete Tomcat image: `docker rmi tomcat`

`docker rm -f <container-id>`

`docker rmi tomcat`

```
ubuntu@ip-172-31-10-98:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	0e760fd9bc48	25.9kB	9.49kB	U
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
ubuntu@ip-172-31-10-98:~$ docker rmi 247650787b97
Untagged: tomcat:latest
Deleted: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
ubuntu@ip-172-31-10-98:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	0e760fd9bc48	25.9kB	9.49kB	U
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
ubuntu@ip-172-31-10-98:~$
```

6. Inspect the Jenkins image, SonarQube image.

Inspect Jenkins Image : docker inspect jenkins/jenkins:lts

```
ubuntu@ip-172-31-10-98:~$ docker inspect jenkins/jenkins:lts
[
  {
    "Id": "sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5",
    "RepoTags": [
      "jenkins/jenkins:lts"
    ],
    "RepoDigests": [
      "jenkins/jenkins@sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-05-13T17:19:33.598838948Z",
    "Config": {
      "User": "jenkins",
      "ExposedPorts": {
        "50000/tcp": {},
        "8080/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "JENKINS_HOME=/var/jenkins_home",
        "JENKINS_SLAVE_AGENT_PORT=50000",
        "REF=/usr/share/jenkins/ref"
      ]
    }
  }
]
```

Inspect SonarQube Image : docker inspect sonarqube

```
ubuntu@ip-172-31-10-98:~$ docker inspect sonarqube
[
  {
    "Id": "sha256:35bedac3f40cda75969890da59b17d57770844fe6ef659206c678a8e00921c7",
    "RepoTags": [
      "sonarqube:latest"
    ],
    "RepoDigests": [
      "sonarqube@sha256:35bedac3f40cda75969890da59b17d57770844fe6ef659206c678a8e00921c7"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-05-11T22:13:40.873390969Z",
    "Config": {
      "User": "sonarqube",
      "ExposedPorts": {
        "9000/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "JAVA_HOME=/opt/java/openjdk",
        "LANG=en_US.UTF-8",
        "LANGUAGE=en_US:en",
        "LC_ALL=en_US.UTF-8",
        "JAVA_VERSION=jdk-25.0.3+9",
      ]
    }
  }
]
```

i-08ba2672d4332489d (docker-test)

7. Run Jenkins container and run one sample job.

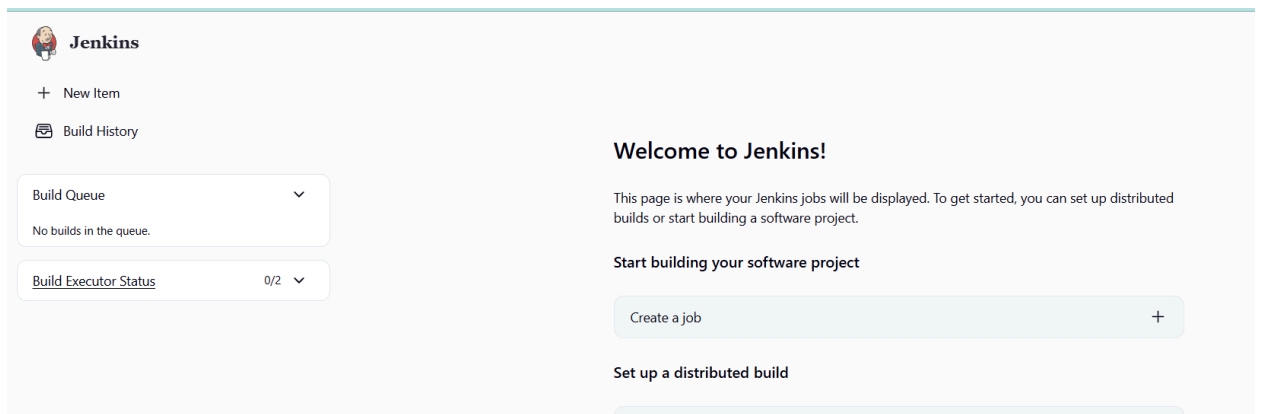
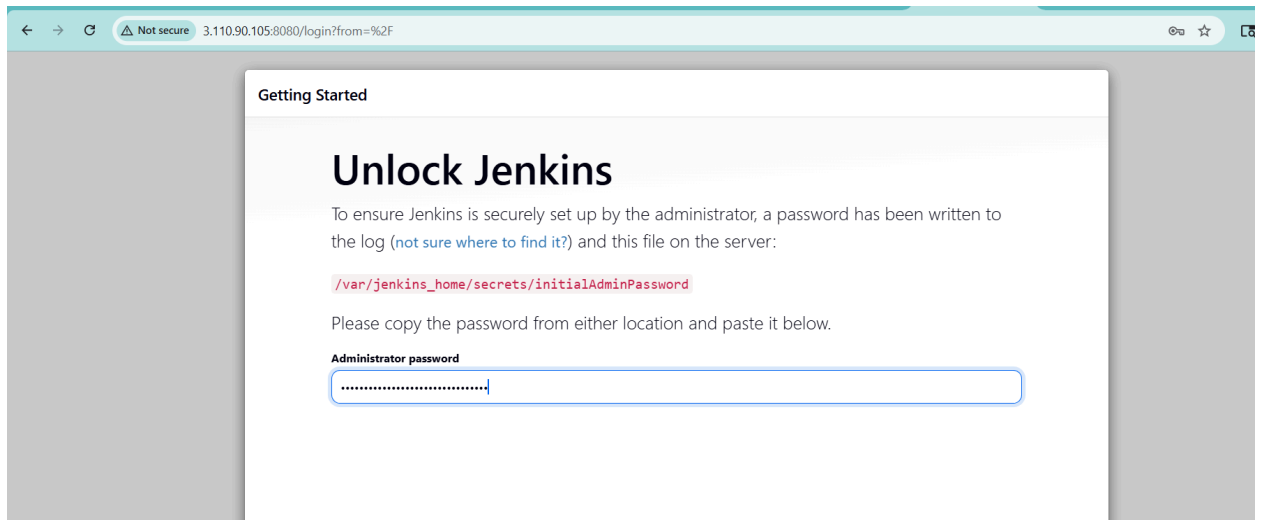
Run Jenkins Container

```
ubuntu@ip-172-31-10-98:~$ docker run -d \
--name jenkins \
-p 8080:8080 \
-p 50000:50000 \
jenkins/jenkins:lts
ec8124ebbbb394dfda8bdeeccd0c76fb278a61606de8ccc4d7af586cc316da19
ubuntu@ip-172-31-10-98:~$ docker ps
```

Check container:

```
ubuntu@ip-172-31-10-98:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
ec8124ebbbb3   jenkins/jenkins:lts   "/usr/bin/tini -- /u..." 12 seconds ago Up 12 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
9dddb9bd4569   nginx          "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  0.0.0.0:81->80/tcp, [::]:81->80/tcp
ubuntu@ip-172-31-10-98:~$
```

docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
Open Jenkins : ip:8080



Create Sample Jenkins Job :

Jenkins / All / New Item

New Item

Enter an item name

sample-job

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Build step:

Jenkins / sample-job / Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell

Command

See the list of available environment variables

echo "Hello Jenkins"
date
hostname

Advanced

+ Add build step

Post-build Actions

Save & build now

Check the console output:



Jenkins

sample-job / #1



Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Console

Download

Copy

View as plain text

```
Started by user bhargav radharapu
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/sample-job
[sample-job] $ /bin/sh -xe /tmp/jenkins10964721977034230961.sh
+ echo Hello Jenkins
Hello Jenkins
+ date
Sat May 30 18:22:59 UTC 2026
+ hostname
ec8124ebbbb3
Finished: SUCCESS
```